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TITLE PLACEHOLDER

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Abstract

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ABSTRACT PLACEHOLDER

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Introduction

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Methods

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Our experimental protocol was approved by the Committee for the Protection of Human Subjects

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at Dartmouth College. We obtained written informed consent from all participants prior to the

11

start of both experiment sessions.

Participants

We enrolled a total of 70 Dartmouth undergraduate students in our study. Participants received optional course credit for enrolling and were pre-screened to ensure they had not previously seen any episode of either TV show viewed during the experiment. No statistical method was used to predetermine sample size. Prior to data processing and analysis, we excluded a total of 17 participants from our dataset: 4 encountered technical issues with the experiment, 2 self-reported difficulty with the task due to illness, 4 spoke for less than 10 minutes during the immediate recall phase, and 7 did not return for the second experiment session. The data we analyzed comprised the remaining 53 participants.

At the start of the experiment, we asked each participant to complete a demographic survey that included questions about their age, sex, race, and ethnicity. Participants' ages ranged from 18 to 22 years (mean: 18.91 years; standard deviation: 1.01 years). A total of 41 participants reported their sex as female and 12 participants reported their sex as male. Participants reported their races as White (34 participants), Asian (16 participants), Black or African American (3 participants), American Indian or Alaska Native (1 participant), and other (1 participant). (Note that some participants selected multiple racial categories.) A total of 52 participants reported their ethnicity as "Not Hispanic or Latino" and 1 participant reported their ethnicity as "Hispanic or Latino."

We also asked participants their native spoken language; whether they had any hearing, speech, or color vision impairments; and about any current medications or recent injuries that could impact attention or memory. All participants reported that their native spoken language was English, that they had no hearing or speech impairments, and that they had normal color vision. A total of 52 participants reported no attention- or memory-relevant medications or injuries. One participant reported sustaining a concussion more than 3 years prior to the experiment and was not excluded from our dataset.

Additionally, before each of the two experiment sessions, we asked participants about their sleep, coffee consumption, and level of alertness. Participants reported having slept between 5 and 11 hours the night prior to session 1 (mean: 7.37 hours; standard deviation: 1.45 hours) and

39 between 5 and 11 hours the night prior to session 2 (mean: 7.39 hours; standard deviation: 1.23
40 hours). They reported having consumed between 0 and 3 cups of coffee on the day of session
41 1 (mean: 0.43 cups; standard deviation: 0.72 cups) and between 0 and 3 cups of coffee on the
42 day of session 2 (mean: 0.48 cups; standard deviation: 0.68 cups). Participants reported their
43 current levels of alertness on a 5-point Likert scale. At the start of session 1, 1 participant reported
44 feeling "Very sluggish," 14 reported feeling "A little sluggish," 18 reported feeling "Neutral," 15
45 reported feeling "A little alert," and 5 reported feeling "Very alert." At the start of session 2, 3
46 participants reported feeling "Very sluggish," 9 reported feeling "A little sluggish," 23 reported
47 feeling "Neutral," 17 reported feeling "A little alert," and 3 reported feeling "Very alert."

48 @jeremy: do we want to report class years and majors? And/or do we want to report post-experiment
49 survey responses (how engaging did you find the episode, how easy/difficult was it to follow, how well do you
50 think you recalled it, how well do you think you learned characters' names)?

51 Experiment

52 The experiment took place over two sessions lasting approximately one hour each, separated by
53 6–8 days (mean: 6.76 days; standard deviation: 0.73 days). Prior to the start of the experiment,
54 participants were assigned to one of two groups (A or B), alternating based on enrollment order.
55 During session 1, all participants first viewed the pilot episode of the FX series *Atlanta* ("The Big
56 Bang"; duration: 24 min 26 s). Immediately after viewing, participants were given up to one hour
57 to recount what happened in the episode to the best of their abilities. Following Chen et al. (2017),
58 participants were instructed to describe the episode in as much detail as possible and to try to
59 speak for a minimum of 10 minutes. They were told to try to recount the episode's events in their
60 original order, but that they were free to go back and describe earlier events should they realize
61 they had missed something. After the immediate recall phase, participants were then given up to
62 5 minutes to predict what they thought would happen in the show's next episode.

63 Upon returning for session 2, all participants were again given up to one hour to recount the
64 *Atlanta* episode they had viewed the week prior. Following the delayed recall phase, participants in
65 Group A then viewed the second episode of *Atlanta* ("Streets on Lock"; duration: 21 min 57 s) while

participants in Group *B* instead viewed the pilot episode of the FOX series *Arrested Development* (“Pilot”; duration: 20 min 37 s). Both groups were then given up to one hour to recount the just-watched episode in the same manner and given the same instructions as in session 1. We implemented the experiment using the psiTurk platform (Gureckis et al., 2016) and jsPsych library (de Leeuw, 2015), with a custom jsPsych plugin we developed in prior work for recording verbal recall audio (Ziman et al., 2018). Here we analyze participants’ immediate and delayed recalls of the *Atlanta* episode viewed in session 1.

Stimulus annotation and recall transcription

To capture the evolving semantic content of the *Atlanta* episode, we created detailed text annotations of the on-screen content during each individual camera shot throughout the episode (mean annotation duration: 3.05 s; standard deviation: 1.80 s). Specifically, we used ANVIL (Kipp, 2001) to record, for each camera shot, a set of semantically relevant features. These included a brief description of physical occurrences and characters’ explicit actions, a brief description of characters’ internal states (e.g., emotions, motivations) and other implicit content, the names of all characters appearing on screen, a transcription of any dialogue, the name(s) of the character(s) speaking, the shot’s setting, whether the shot took place indoors or outdoors, and whether or not there was music playing during the shot. This feature set was adapted from a similar annotation corpus created by Chen et al. (2017), which we used in developing our trajectory-analysis framework in prior work (Heusser et al., 2021).

We then transcribed each participant’s immediate and delayed verbal recall of the *Atlanta* episode. Our transcription process consisted of three steps: an initial manual transcription, a two-pass prose-formatting conversion with a large language model, and a final manual cleanup and correction step. First, we manually transcribed each recall audio file collected during the experiment. These initial transcripts reflected participants’ spoken recalls verbatim, without any punctuation or additional formatting, and including disfluencies, stuttered repetitions, and other artifacts common to unrehearsed speech. However, since transformer models (such as we used to obtain text embeddings of participants’ recalls) are trained almost exclusively on written text,

93 structural conventions like sentence breaks, punctuation, and capitalization provide an important
94 signal these models leverage to contextual other tokens (e.g., words) and construct a representation
95 of the text’s overall meaning. Meanwhile, speech-native phenomena such as stuttered repetitions
96 and filler words (“like”, “um”, etc.) that are comparatively rare in written text contribute out-of-
97 distribution noise that can be expected to degrade embedding fidelity, and prior work has shown
98 that automated removal of these phenomena improves performance on downstream tasks that
99 assume clean, written-style input (Chen et al., 2022; Wang et al., 2010). We therefore used a large
100 language model (Qwen3.5 122b; Qwen Team, 2026) to convert participants’ recall transcripts to a
101 structure and style more typical of written text. We used the `langchain_dartmouth` Python package
102 (Stone et al., 2025) to implement this conversion as two sequential passes over each transcript. In the
103 first pass, we prompted the model to split the transcript into sentences at natural breaking points,
104 add punctuation including quotes around reported speech, remove clear non-word disfluencies
105 (“um,” “uh,” “hmm,” etc.), collapse stuttered verbatim repetitions, and mark self-corrections and
106 restarts with em-dashes. In the second pass, we prompted the model to remove all instances of
107 “like” used as a filler word. We removed filler “like” instances in a separate pass because of their
108 high frequency of occurrence in the transcripts and the additional effort to required to distinguish
109 them from meaningful uses of “like.” The full texts of our prompts for both passes can be found in
110 *Supplementary methods*. Finally, we manually reviewed all LLM-formatted transcripts to identify
111 and correct any errors, ensure sentence breaks were appropriate and consistent across participants,
112 confirm that no content was improperly added to or removed from the transcripts, and resolve
113 any ambiguities using the prosody of participants’ original recall audio.

114 **Analysis**

115 **Statistics**

116 Wilcoxon tests were one-sided because our alternative hypothesis was directional (delayed matches
117 own immediate *better than* the episode, others’ immediates, etc.).

118 **Constructing episode and recall trajectories**

119 **Event-segmentation**

120 **2D path embedding**

121

122 **Creating knowledge and learning map visualizations**

123 **Data availability**

124 All of the data analyzed in this manuscript may be found at [https://github.com/ContextLab/memory-](https://github.com/ContextLab/memory-dynamics)
125 [dynamics](https://github.com/ContextLab/memory-dynamics).

126 **Code availability**

127 All of the code for running our experiment and carrying out the analyses may be found at
128 <https://github.com/ContextLab/memory-dynamics>.

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160 **Acknowledgements**

161 INDIVIDUAL ACKNOWLEDGEMENTS PLACEHOLDER. Our work was supported in part by
162 NSF CAREER Award Number 2145172 to J.R.M. The content is solely the responsibility of the
163 authors and does not necessarily represent the official views of our supporting organizations.
164 The funders had no role in study design, data collection and analysis, decision to publish, or
165 preparation of the manuscript.

166 **Author contributions**

167 **Competing interests**

168 The authors declare no competing interests.